Speed of light in a vacuum:

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$$

ε_0 = electric permitivity of free space, $= 8.85 \cdot 10^{-12} \frac{F}{m}$

measures how easily electric field can form and pass through free space

$\mu_0 = 1.26 \cdot 10^{-6} \frac{N}{A^2}$ = magnetic permeability of free space.

measures how easily a magnetic field can propagate

$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$ comes from Maxwell's equations. Light is an <u>electromagnetic wave</u>, meaning it consists of oscillating E and B fields perpendicular to each other and the direction of wave propagation.

<u>E field by a point charge q:</u>

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \quad \text{OR} \quad \frac{k\,q}{r^2} \quad \text{where } k = \frac{1}{4\pi\varepsilon_0}$$

Consider a charged particle moving at a fraction of c, so

$v_q = \xi c, \; 0 < \xi < 1$

with reference to frame S.

Let S' be a ref. frame moving with the charged particle. So,

$$\frac{dx'}{dt'} = v_q' = 0$$

S' origin is q's pos.

<u>Incorrect assumption:</u>

$$\frac{dx'}{dt'} = \frac{dx}{dt} - v_q$$

① Assume q is an electron. Find eqn. for E as a function of r' from q in frame S'!

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r')^2}, \quad q = -e$$

where e = charge of one proton $= 1.602 \cdot 10^{-19}$

<u>Galilean velocity transformation:</u>

- objects can "catch up" with light
- light in a vacuum appears slower than c.

② Plot c', <u>incorrect</u> speed of light in S' according to Galilean transformation as a function of ξ.

$$c' = c - v_q, \quad v_q = \xi c, \quad 0 < \xi < 1$$
$$= c - \xi c = c(1-\xi)$$

③ Contradiction! If $c' < c$, eqn. 1 is false, or μ_0 and/or ε_0 must change. Assume eqn. 1 holds, μ_0 is constant, and ε_0 varies.

Plot (a) ε_0' as a function of ξ and (b) E field E' at r' = 1μm as a function of ξ.

(a): $c' = \frac{1}{\sqrt{\varepsilon_0' \mu_0}}$ $\quad c' = c(1-\xi)$

$$c(1-\xi) = \frac{1}{\sqrt{\varepsilon_0' \mu_0}}$$

$$(\mu_0)^{\frac{1}{2}} c(1-\xi) = \frac{1}{\sqrt{\varepsilon_0'}}$$

$$\sqrt{\varepsilon_0'} = \frac{1}{\sqrt{\mu_0}\, c(1-\xi)}$$

$$\varepsilon_0' = \frac{1}{(\sqrt{\mu_0}\, c(1-\xi))^2}$$

(b) $E' = \frac{1}{4\pi\varepsilon_0'}\left(\frac{q}{r'^2}\right)$

$$= \frac{(\sqrt{\mu_0}\, c(1-\xi))^2}{4\pi}\left(\frac{q}{r^2}\right)$$

④ Assume $\xi = 0.5$. Plot E'(r') by std. equation (eqn 2) and from problem 3, where ε_0' is a function of ξ.

std: $E'(r') = \frac{1}{4\pi\varepsilon_0'}\left(\frac{q}{r'^2}\right)$

$\xi = 0.5$: $E'(r') = \frac{(\sqrt{\mu_0}\, c(0.5))^2}{4\pi}\frac{q}{r^2}$

⑤ Vary μ_0' instead of ε_0'?

$c' = \frac{1}{\sqrt{\varepsilon_0 \mu_0'}}$ $\quad c' = c(1-\xi)$

Let a current loop (I = 1A) be at rest in S'. Magnetic field strength at center of loop in S is $B = \frac{\mu_0 I}{2R}$. R is radius, R = 1m.

Assume E' = E, B' = B, c' depends on ξ as in problem 2. Plot μ_0 vs. ξ.

$$\frac{1}{\sqrt{\varepsilon_0 \mu_0'}} = c(1-\xi)$$

$$\frac{1}{\sqrt{\varepsilon_0}\, c(1-\xi)} = \sqrt{\mu_0'}$$

$$\mu_0' = \frac{1}{(\sqrt{\varepsilon_0}\, c(1-\xi))^2}$$

⑥ Any value of ξ for which E' = E and B' = B? (Given $c' = c(1-\xi)$)

$$\left.\begin{array}{l} E' = E \Rightarrow \varepsilon_0' = \varepsilon_0 \\ B' = B \Rightarrow \mu_0' = \mu_0 \end{array}\right\} \text{both true}$$

$$\varepsilon_0' = \frac{1}{(\sqrt{\mu_0}\, c(1-\xi))^2}$$

$$\mu_0 = \frac{1}{(\sqrt{\varepsilon_0'}\, c(1-\xi))^2}$$

$$\varepsilon_0' = \frac{1}{(\sqrt{\mu_0}\, c(1-\xi))^2}$$